CONG ZHANG

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INTRODUCTION

I am currently an Assistant Research Fellow at the State Key Laboratory of Cryptography and Digital Economy Security, Tsinghua University. Previously, I was a Postdoctoral Fellow at the Institute for Advanced Study, Tsinghua University, working under the supervision of Prof. Xiaoyun Wang. I earned my Ph.D. from the Institute of Information Engineering, Chinese Academy of Sciences in 2023, where I was advised by Prof. Dongdai Lin. Prior to my doctoral studies, I completed my Bachelor's degree in Mathematics at Shandong University in 2018.

My main research focus is on cryptographic protocols for secure computation. I am interested in both theoretical and practical aspects of secure computation techniques. More specifically:

- **General Multi-Party Computation Protocols**, especially Garbled Circuit (GC) and Secret Sharing based protocols.
- **Specific Protocols**, especially Private Set Operations (PSO), Private Information Retrival (PIR) etc.

EXPERIENCE

State Key Laboratory of Cryptography and Digital Economy Security, Tsinghua University — 2026.09- Now
Assistant researcher.

Institute for Advanced Study, Tsinghua University — 2024.01- 2026.08
Postdoctoral researcher. Advisor: Prof. Xiaoyun Wang

Alibaba Group — 2023.07 - 2023.10
Research Intern, with Dr. Weiran Liu and Dr. Liqiang Peng

EDUCATION

Ph.D., Institute of Information Engineering, Chinese Academy of Sciences, Beijing, China — 2018 - 2023

- Research directions: Cryptographic Protocols for Secure Computation
- Thesis: Research on Secure Multiparty Computation for Privacy Protection

Bachelor, Shandong University, Shandong, China — 2014 - 2018

PUBLICATIONS

- **Cong Zhang**, Yu Chen, Yujie Bai, Yang Cao, Shuaishuai Li, Juntong Lin, Anyu Wang, and Xiaoyun Wang. Breaking the Curse of Dimensionality: Fuzzy PSI without $O((\log \delta)^d)$ Overhead. In Proceedings of the 2026 ACM SIGSAC Conference on Computer and Communications Security, CCS 2026.
- Shuaishuai Li, **Cong Zhang**, Juntong Lin, Anyu Wang, Xiaoyun Wang. Information-Theoretic SFE and PFE with Reduced Communication. In the 32nd International Conference on the Theory and Application of Cryptology and Information Security, ASIACRYPT 2026.
- Juntong Lin, Shuaishuai Li, **Cong Zhang**, Zhiyuan Qiu, Qidi You, Anyu Wang and Xiaoyun Wang. Private Set Union under Hidden Feature Overlap for Security Intelligence Ingestion. In the 22nd International Conference on Information Security and Cryptology, Inscrypt 2026.
- Shuaishuai Li, **Cong Zhang**, Anyu Wang and Xiaoyun Wang. Private Function Evaluation with Linear Complexity. In 46th Annual International Cryptology Conference, CRYPTO 2026.
- **Cong Zhang**, Liqiang Peng, Weiran Liu, Shuaishuai Li, Meng Hao, Lei Zhang, and Dongdai Lin. Charge Your Client: Payable Secure Computation and Its Applications. In IEEE Transactions on Information Forensics and Security, TIFS 2025.

- Shuaishuai Li, Liqiang Peng, Weiran Liu, **Cong Zhang**, Zhen Gu, and Dongdai Lin. BitBatSPIR: Efficient Batch Symmetric Private Information Retrieval from PSI. In IEEE Transactions on Dependable and Secure Computing, TDSC 2025.
- Minglang Dong, Yu Chen, **Cong Zhang**, Yujie Bai, and Yang Cao. Multi-Party Private Set Operations from Predicative Zero-Sharing. In Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security, CCS 2025.
- Minglang Dong, **Cong Zhang**, Yujie Bai, and Yu Chen. Efficient Multi-Party Private Set Union Without Non-Collusion Assumptions. In 34th USENIX Security Symposium 2025.
- Binbin Tu, Yujie Bai, **Cong Zhang**, Yang Cao, and Yu Chen. Fast Enhanced Private Set Union in the Balanced and Unbalanced Scenarios. In 34th USENIX Security Symposium 2025.
- Meng Hao, Weiran Liu, Liqiang Peng, **Cong Zhang**, Pengfei Wu, Lei Zhang, Hongwei Li, and Robert H. Deng. Practical Keyword Private Information Retrieval from Key-to-Index Mappings. In 34th USENIX Security Symposium 2025.
- **Cong Zhang**, Yu Chen, Weiran Liu, Liqiang Peng, Meng Hao, Anyu Wang and Xiaoyun Wang. Unbalanced Private Set Union with Reduced Computation and Communication. In Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security, CCS 2024.
- Shuaishuai Li, **Cong Zhang**, and Dongdai Lin. Secure Multiparty Computation with Lazy Sharing. In Proceedings of the 2024 ACM SIGSAC Conference on Computer and Communications Security, CCS 2024.
- Yu Chen, Min Zhang, **Cong Zhang**, Minglang Dong, and Weiran Liu. Private Set Operations from Multi-Query Reverse Private Membership Test. In Public Key Cryptography, PKC 2024.
- Meng Hao, Weiran Liu, Liqiang Peng, Hongwei Li, **Cong Zhang**, Hanxiao Chen, and Tianwei Zhang. Unbalanced Circuit-PSI from Oblivious Key-Value Retrieval. In 33rd USENIX Security Symposium 2024.
- Xiangling Zhang, **Cong Zhang**, Weiran Liu, and Yu Chen. A Survey on Key Data Structures in Private Set Operation. Journal of Cryptologic Research, 2024, 11(2): 263–281. [DOI: 10.13868/j.cnki.jcr.000679]
- **Cong Zhang**, Yu Chen, Weiran Liu, Min Zhang, and Dongdai Lin. Linear Private Set Union from Multi-Query Reverse Private Membership Test. In 32nd USENIX Security Symposium 2023.
- **Cong Zhang**, Weiran Liu, Bolin Ding, and Dondai Lin. Efficient Private Multiset ID Protocols. In the 25th International Conference on Information and Communications Security, ICICS 2023.
- Shuaishuai Li, **Cong Zhang**, and Dongdai Lin. Oblivious Transfer from Rerandomizable PKE. In the 25th International Conference on Information and Communications Security, ICICS 2023.
- Binbin Tu, Yu Chen, Qi Liu, and **Cong Zhang**. Fast Unbalanced Private Set Union from Fully Homomorphic Encryption. In Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security, CCS 2023.
- **Cong Zhang**, Shuaishuai Li, and Dongdai Lin. Amortizing Division and Exponentiation. In the 18th International Conference on Information Security and Cryptology, Inscrypt 2022.
- Rui Zhang, Huan Zou, **Cong Zhang**, Yuting Xiao, and Yang Tao. Distributed Key Generation for SM9-based Systems. In the 16th International Conference on Information Security and Cryptology, Inscrypt 2020.

PROFESSIONAL ACTIVITIES

External Reviewer: Inscrypt (2019,2022,2024,2026), Asiacrypt (2020,2021,2023,2026), ESORICS (2021,2022), PKC 2022, ACM CCS 2022, ICICS 2023, NDSS 2025, CRYPTO 2026.

Program Committe: ACM CCS 2026, NDSS 2027.

SELECTED AWARDS

- 2022 Fincrypto National Competition: Outstanding Winner

- 2023 Director Excellence Award of Institute of Information Engineering, Chinese Academy of Sciences
- 2024 Outstanding Graduates of Institute of Information Engineering, Chinese Academy of Sciences
- 2024 Distinguished Doctoral Dissertation Award of Chinese Association for Cryptologic Research

TALKS

A Simple Introduction to Secure Multi-Party Computation.

- AIMS Conference 2026, Athens, Greece, 2026.07

Fast Enhanced Private Set Union in the Balanced and Unbalanced Scenarios.

- USENIX Security 2025, Seattle, U.S.A., 2025.08

Design of Two-Party Private Set Union Protocols.

- Seminar on Cryptographic Algorithms for Confidential Computing, Institute of Information Engineering, Chinese Academy of Sciences, 2024.11
- Cross Forum of Young Scholars at Institute of Advanced Study, Tsinghua University, 2024.11
- Chinese Association for Cryptologic Research 2025 Youth Forum, Lanzhou, 2025.05

Unbalanced Private Set Union with Reduced Computation and Communication.,

- ACM CCS 2024, Salt Lake City, U.S.A., 2024.10

Secure Multiparty Computation with Lazy Sharing.,

- ACM CCS 2024, Salt Lake City, U.S.A., 2024.10

Efficient Private Multiset ID Protocols.,

- ICICS 2023, Nankai University, Tianjin, China, 2023.11

Linear Private Set Union from Multi-Query Reverse Private Membership Test.,

- USENIX Security 2023, Anaheim, CA, U.S.A., 2023.08

Amortizing Division and Exponentiation.,

- Inscrypt 2022, Online, 2022.12

SELECTED FUNDED PROGECTS

- "Research on Key Cryptographic Protocol in Privacy Computing" from NSFC (No. 62402272), 2025.01 - 2027.12, Total award: 300K RMB.